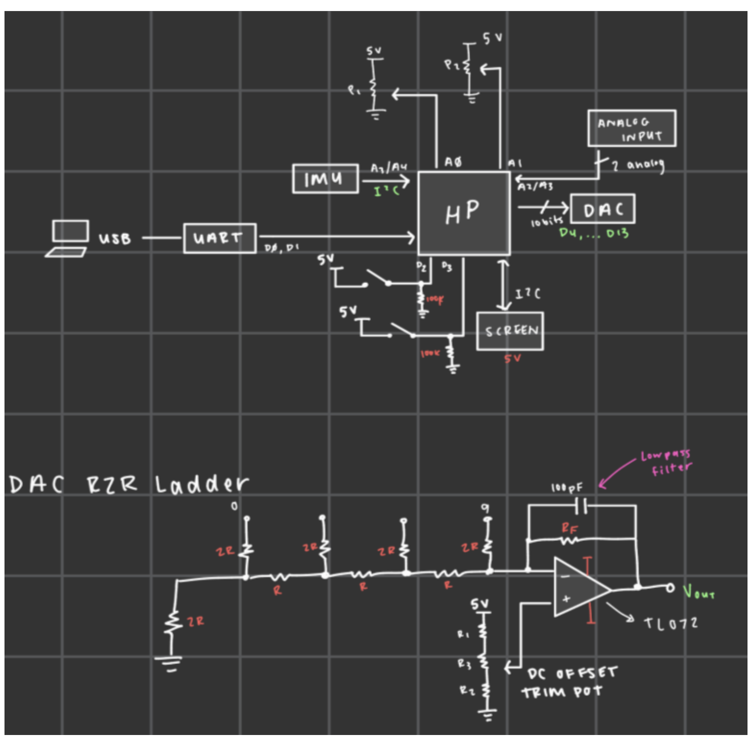
DSP ARDUINO SHIELD v2

INTRODUCTION

The purpose of this project was to design and test a DSP shield for an Arduino UNO with a 10-bit DAC. Using KiCAD for the PCB design, this DSP shield was created for ENCE 3210 Microprocessor Systems to learn PCB design and practice soldering small PCB components.

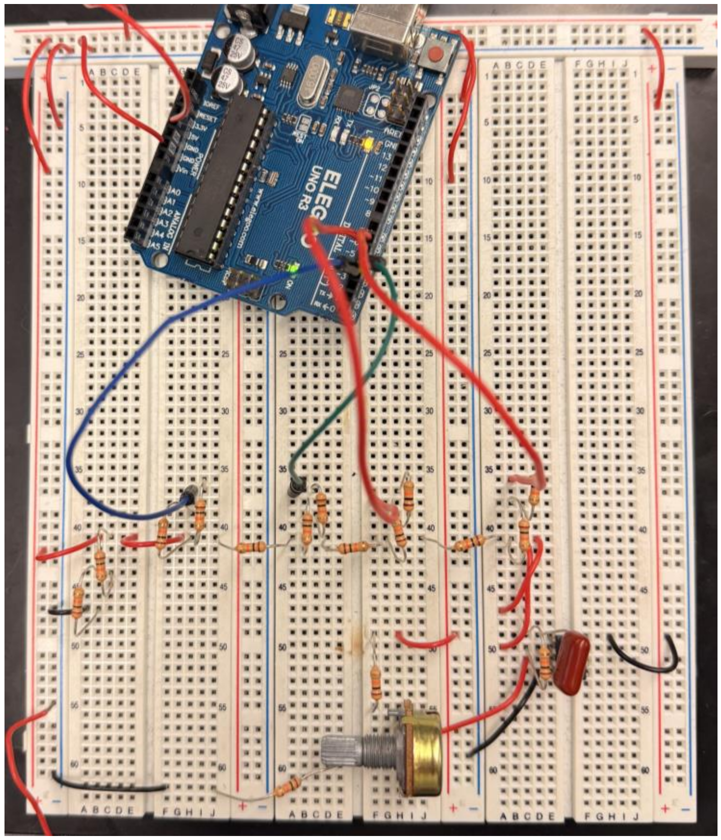
DESIGN

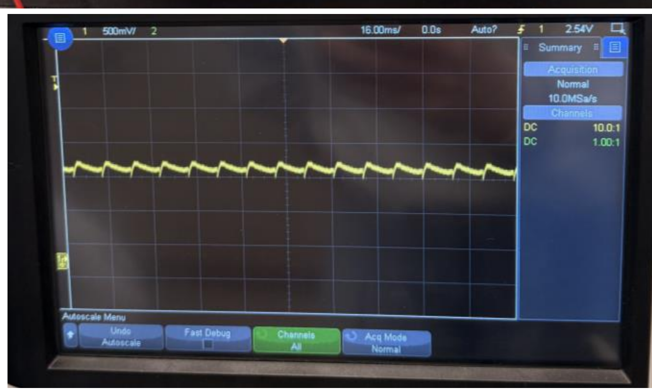
This design was to integrate the following components: 10-bit DAC, OLED 128x32 screen, MPU 6050 IMU, three potentiometers, two buttons, and two analog inputs.



PROTOTYPE

The system was tested before implementation by creating a 4-bit R2R ladder DAC on a breadboard and testing the output on the oscilloscope.



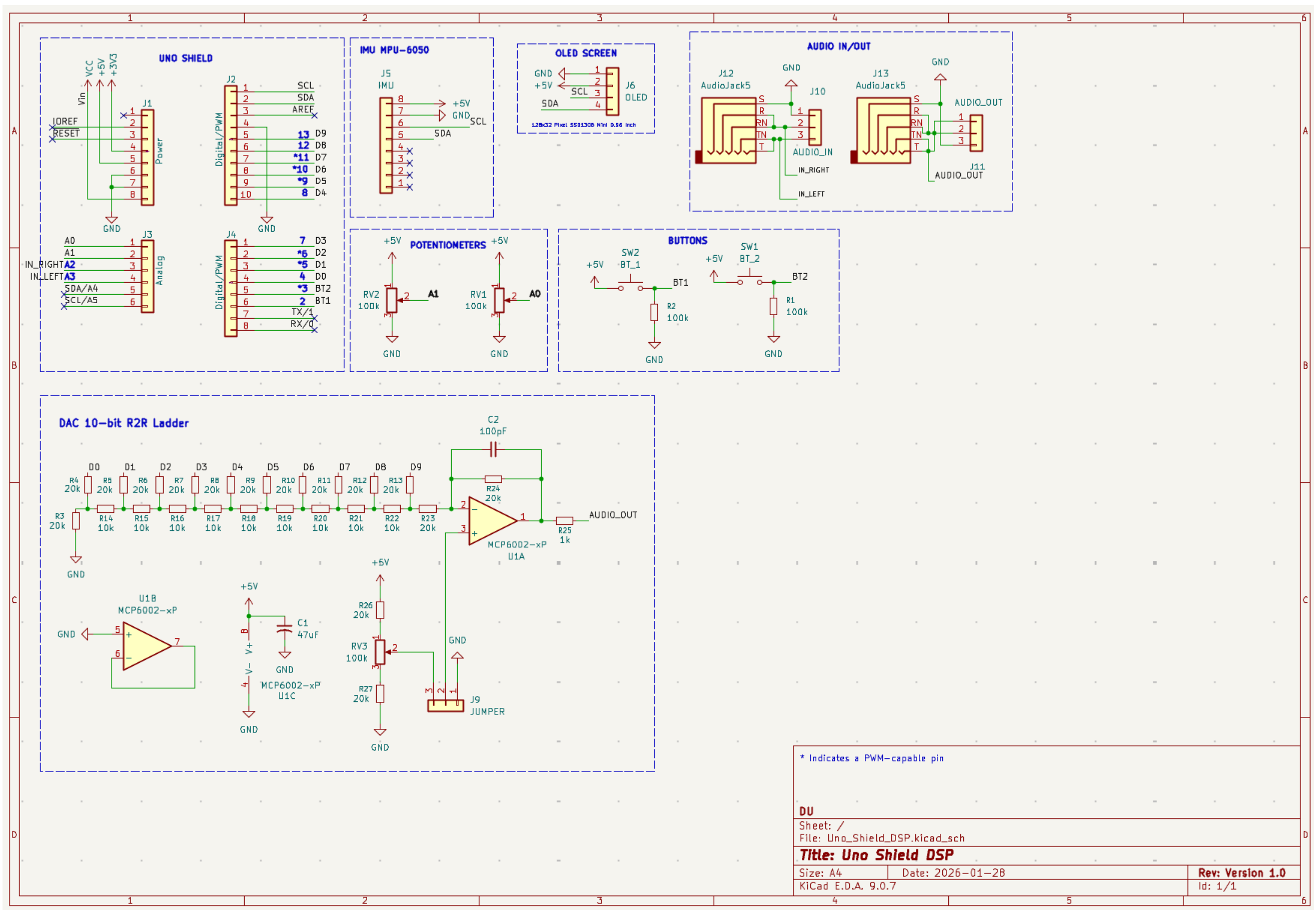


***NOTE: The clipped output was due to incorrect code implementation.**

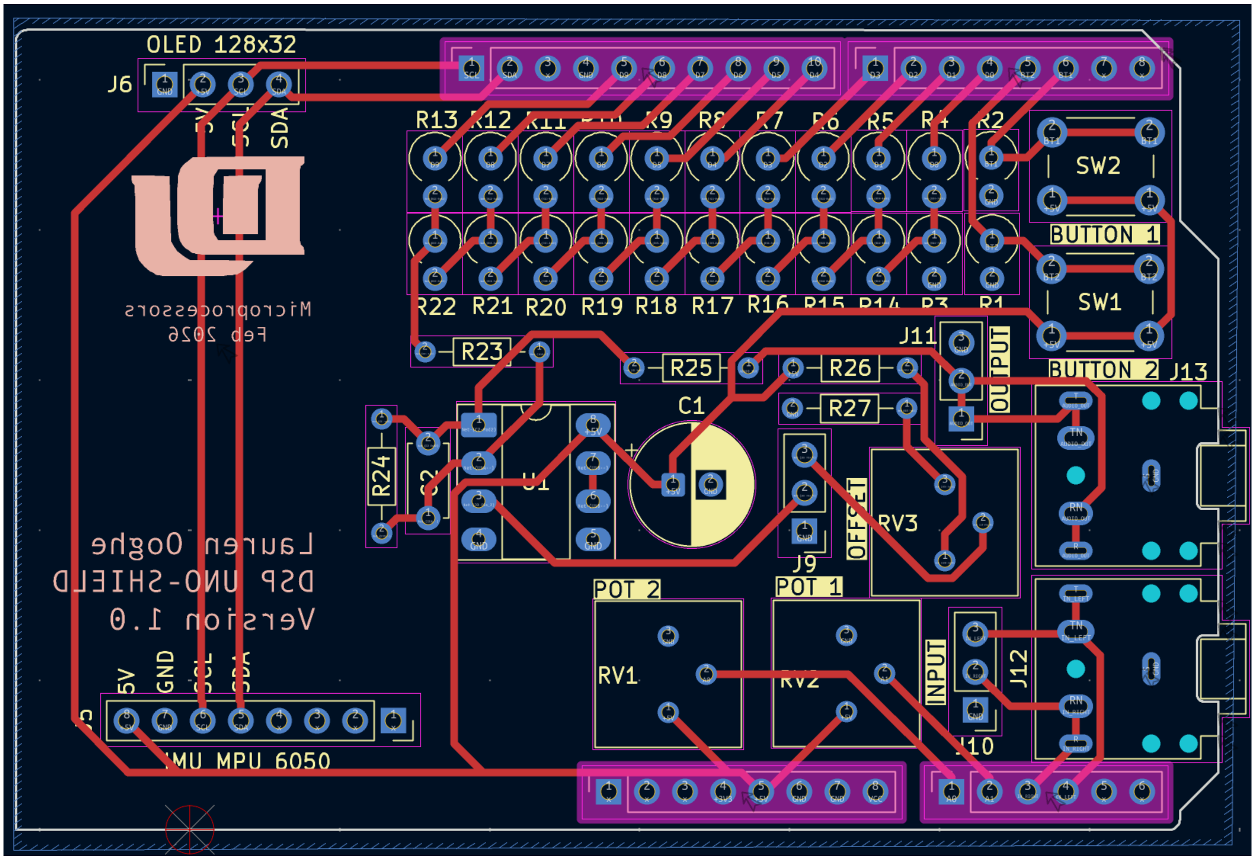
PCB DESIGN

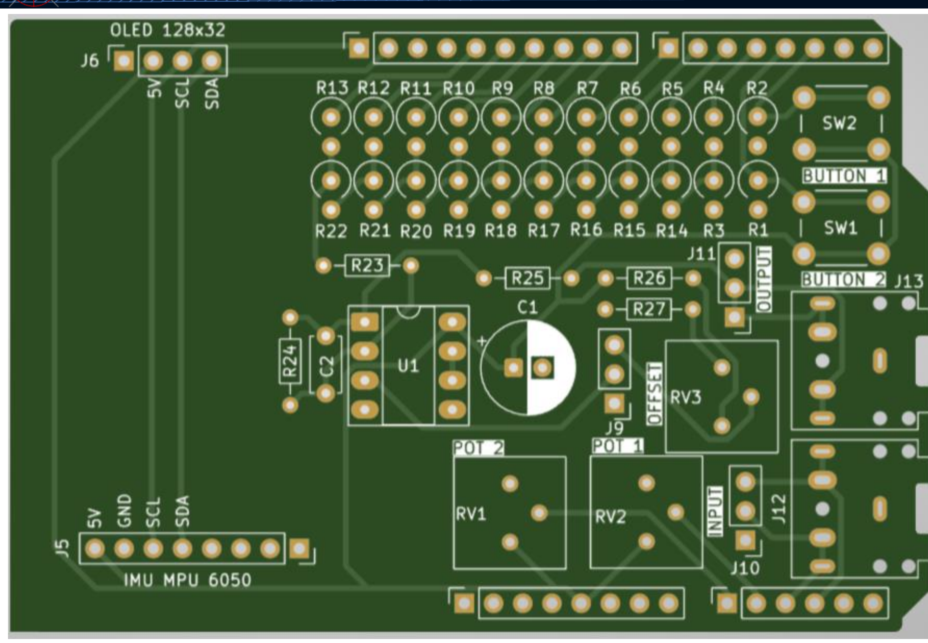
The PCB was designed using KiCAD. This is v2.

Schematics

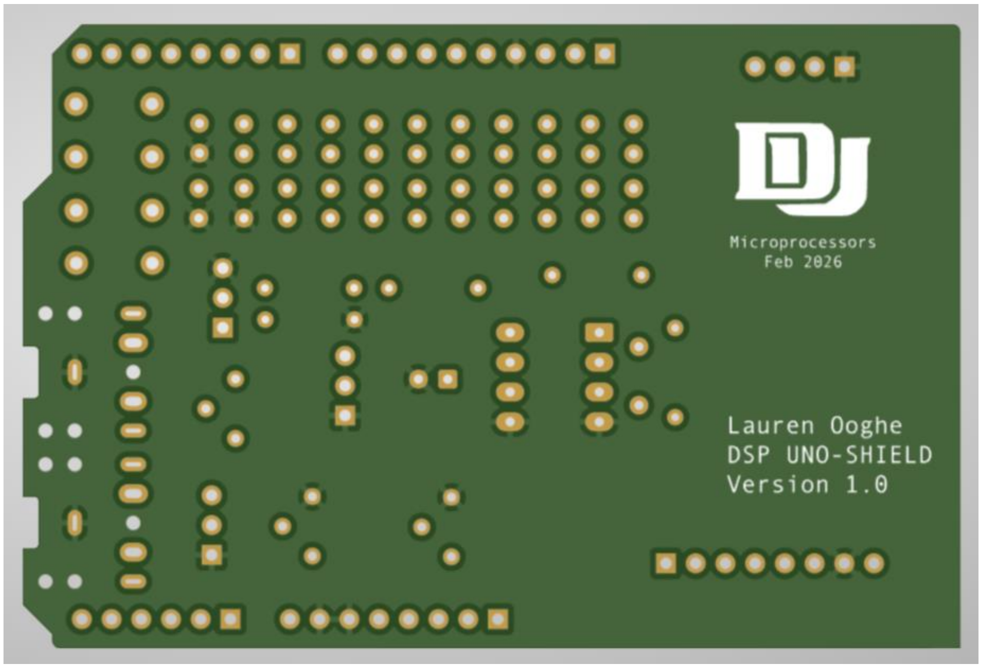


Layout

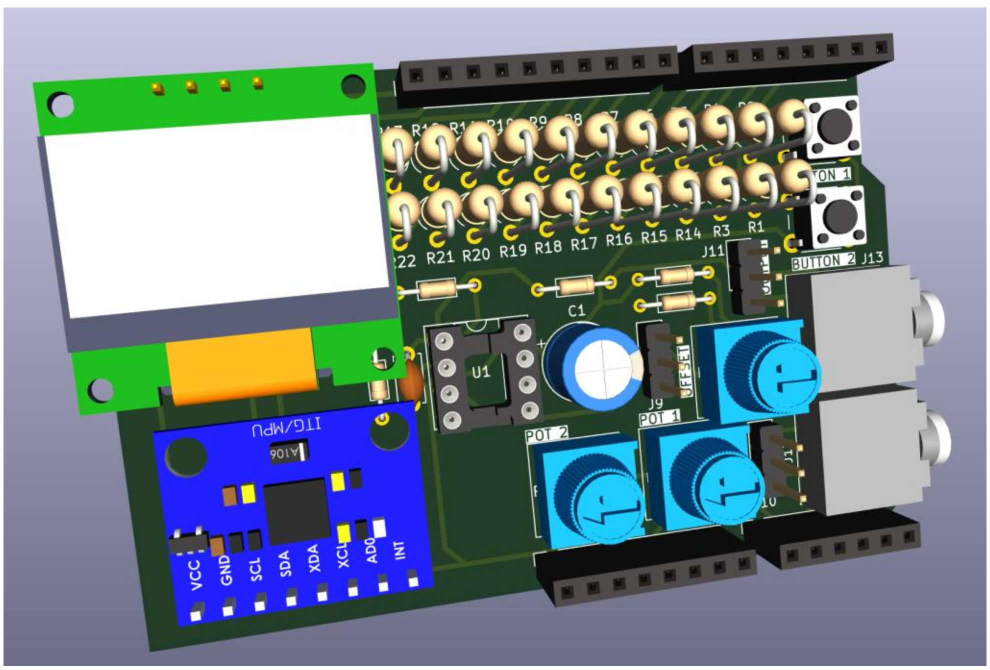




FRONT

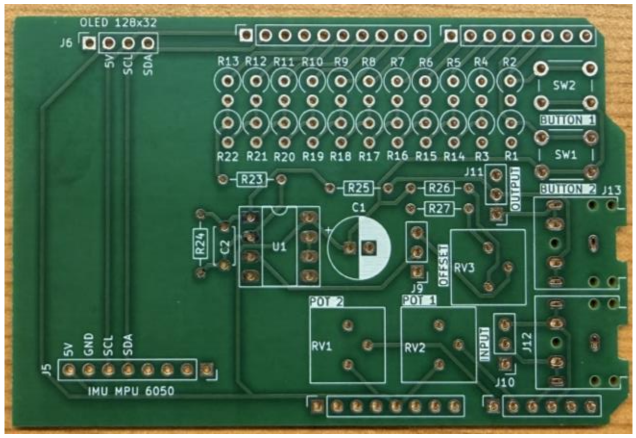

Microprocessors
Feb 2026
Lauren Ooghe
DSP UNO-SHIELD
Version 1.0

BACK

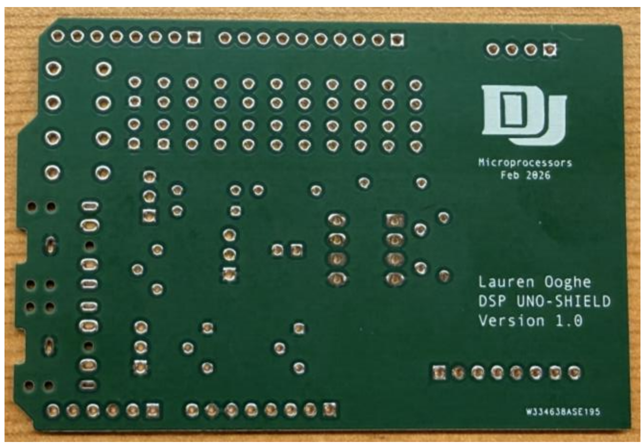

ITG/MPU
VCC
GND
SCL
SDA
XDA
XCL
AD0
INT
U1
C1
POT 2
POT 1
OFFSET
BUTTON 2
J13

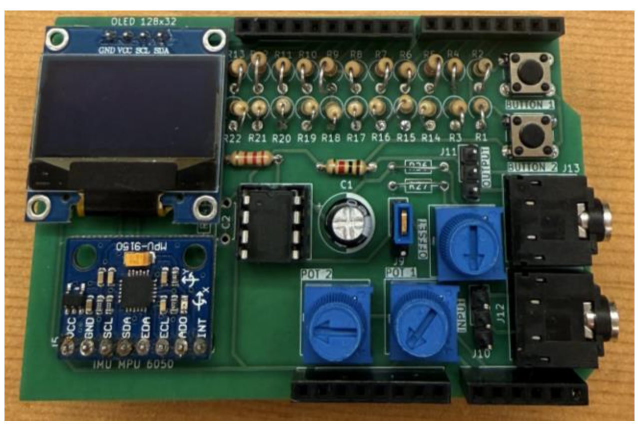
3D VIEW

ASSEMBLY



Boards manufactured from PCBWay



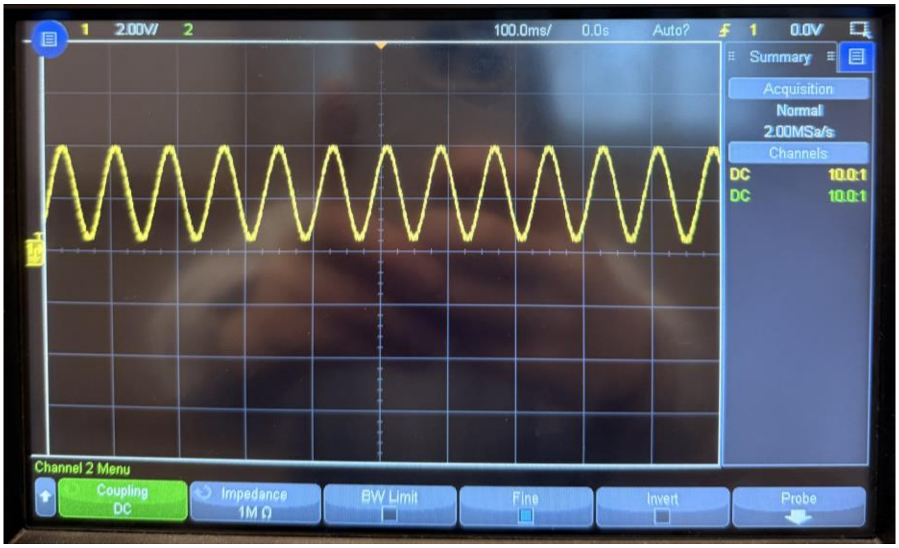


FINAL v2

***NOTE: Capacitor C2 was not used in this design**

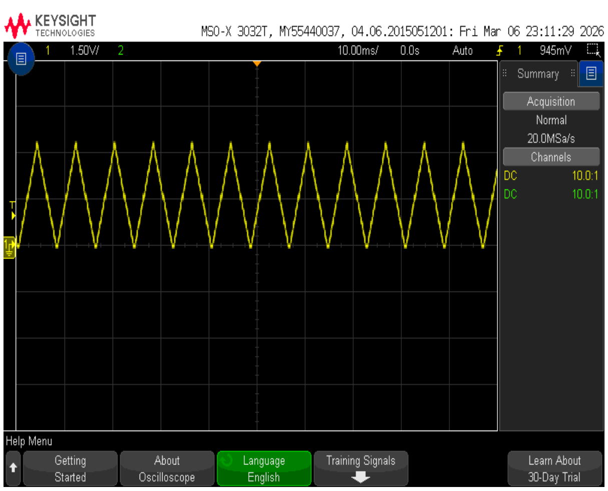
TESTING

The program in this folder named **DAC_ISR_TEST.ino** was used to test a sine wave output.

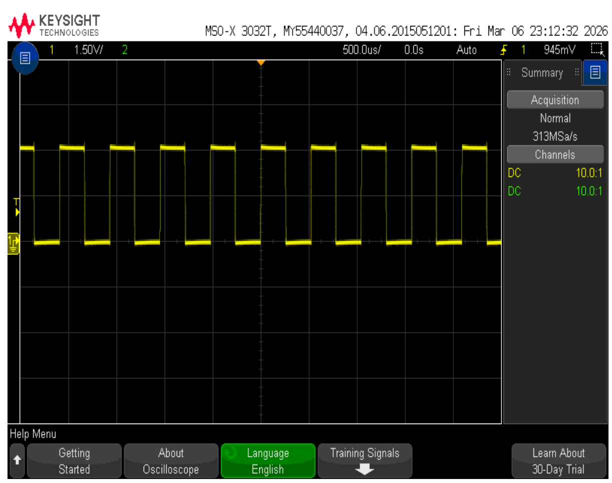


Sine wave output

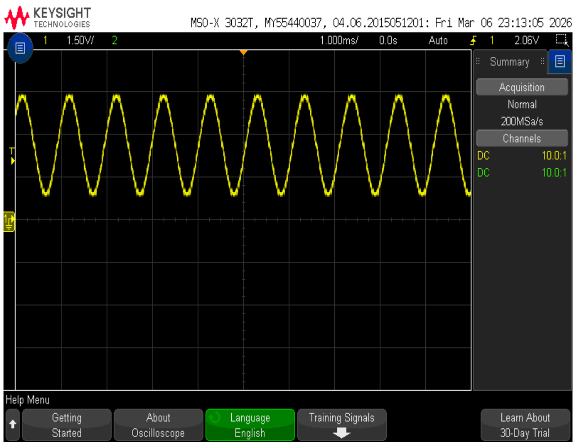
Then the program in this folder named **DSP_SHIELD_ROTATING_MENU.ino** was used to apply different wave forms (triangle, sawtooth, sine, square).



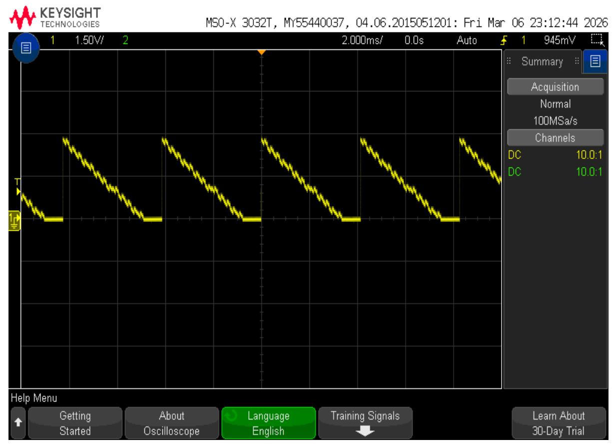
Triangle



Square



Sine



Sawtooth

REFERENCES

All programs used for testing were instructor provided.